Introduction To System Design Seminar

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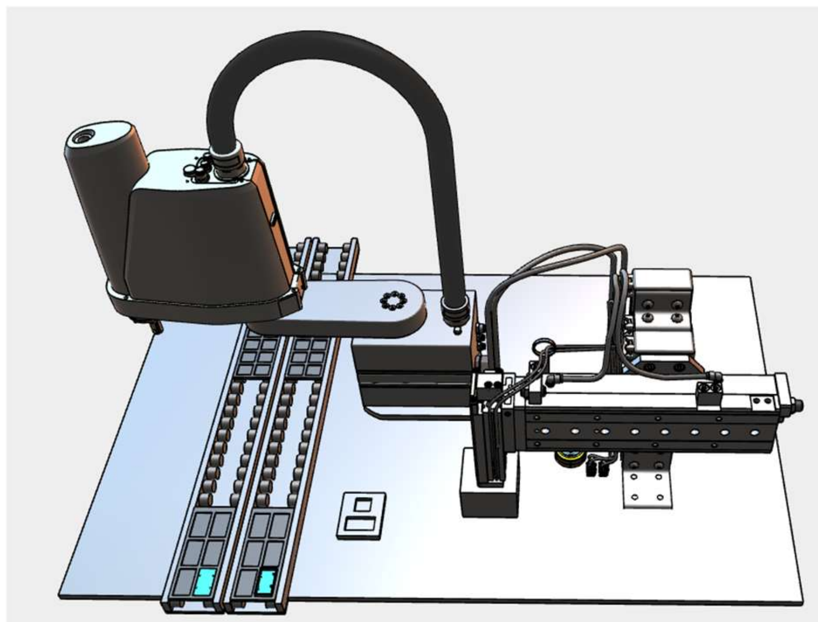
Notes: I created this presentation, the Fusion 360 models, and electrical drawings in Siemens Capital X Panel Design in a week. The slides are an outline; my actual content was much more detailed.

What Is System Design?

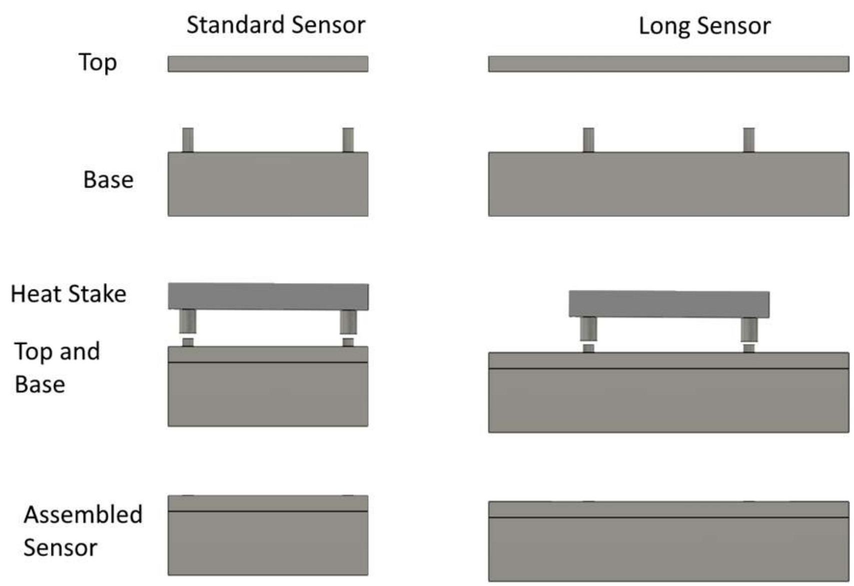
- System Design is defining the architecture and design of the entire system to meet the customer's requirements.
- For an automated machine, we need to understand how the mechanical, electrical, pneumatic, and software parts will interact and come together to satisfy the requirements
- It is critical for everyone involved in the machine's design and production to work together and consider the other parts of the machine.
  - For example, the software engineer cannot fix, using only software, a missing but critically needed sensor.

What does this really mean?

Let’s look at the System Design of a realistic machine, the Heat Stake Assembly Machine (HSAM)



The HSAM’s Purpose

- Uses a heat stake to permanently join the top and bottom of a water-resistant IoT sensor.
- A heat stake uses temperature and pressure to reform the top of a plastic stake into a permanent joint.
- 

Roadmap

- The Big Picture: a high-level look at the requirements and different approaches
- The Details
  - Component Choices: what components we chose and why
  - Software and Communications: details on how the software will work, and how the different controllers will communicate
  - Power and Safety: details on the safety features and electrical schematics
- Summary

The Big Picture

HSAM Design Requirements

1. Safe
2. Meet government and customer regulations and certifications
3. Be reliable, need minimal operator attention, and be easy to maintain
4. Cost effective
    1. Bill of Materials target is under $30,000
5. Needs to be quick to develop – lead customer wants it now
6. Low volume with unpredictable demand
7. Customer will be able to maintain machine
8. Accommodate future add-ons such as cameras

Regulations and Certifications

- Local jurisdictions have safety requirements we have to meet
  - Common US requirements include OSHA, NEC, NFPA 79, and UL508A
    - NEC, NFPA 79, and UL508A are concerned with eliminating fire and shock hazards
    - OSHA regulations are for workplace safety
  - Europe has the comprehensive set of CE requirements
    - There are CE requirements for about everything, including fire and shock hazards, workplace safety, radiated emissions, and electrical immunity
  - China and other countries have their own regulations
  - Customers will also add requirements
    - In the US, requiring UL508A certified panels is common
    - Requiring SEMI S2 and S8 is common in the semiconductor and hard drive industry
      - SEMI S2 contains safety requirements, including worker safety (EMO)
      - SEMI S8 contains ergonomic requirements
  - Many organizations create standards: UL, IEC, ANSI, RIA

Safety Goals and Choices

We want our machines to be safe

- The best way to make the machine safe is to design out, as far as possible, any potential hazards.
  - Eliminate pinch points
  - High voltage electrical systems in locked cabinets
  - Barriers
- Next, mitigate what hazards remain
  - Emergency Stop and/or EMO circuits
  - Short circuit protection
  - Light curtains
  - Safety mats
  - Laser safety scanners
  - Two hand controls
  - Cobots
- Safety must not be cumbersome in use
  - Otherwise, the operators will bypass it – and any safety system can be bypassed



Overall Design Approaches

- Part loading / unloading
  - Manual single part
  - Manual trays
  - Trays on conveyors
- Motion System
  - Pneumatics
    - Force
  - Motorized stages (cartesian)
  - Robot
    - SCARA, articulated, delta
  - Cobot

Electrical Design Approaches

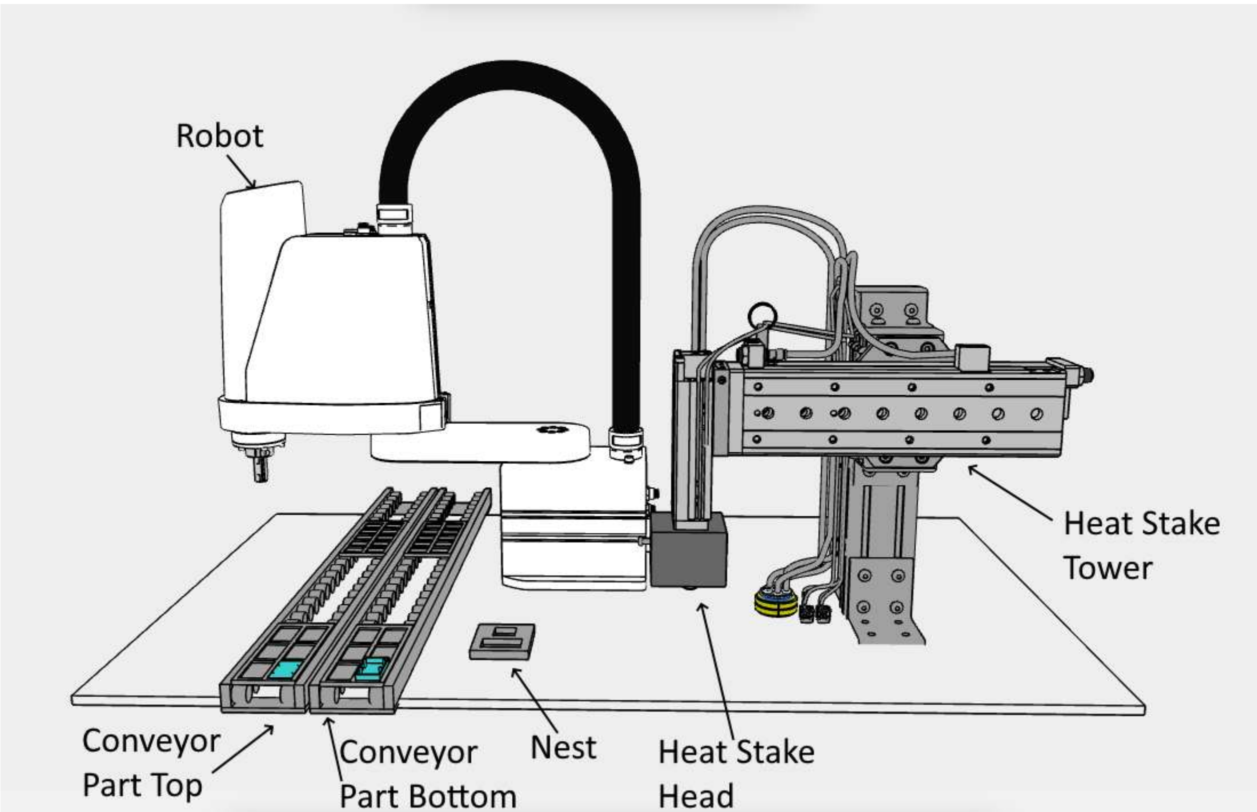
- Wire directly to PLC
  - Not recommended, since the power and ground wires still have to go somewhere
- Use terminal blocks designs for sensors and actuators, such as the Phoenix STIO series.
  - Common and Power levels are bridged, signal wire fed through to PLC
  - Worth the price premium for quicker, more reliable wiring.
- Use custom-designed break out boards
  - Excellent solution for high volume machines with standard configurations
- Connection Technologies
  - Screw terminal
  - Spring clamp
  - Insulation Displacement

How Can Components Communicate?

- Digital Inputs and Outputs
- Analog Inputs and Outputs
  - 0-5V, 0-10V, 20mA, etc
- Serial ports
  - RS232C, RS422, RS485
- USB Ports
  - USB protocol, serial over USB
- Fieldbus
  - DeviceNet, Profibus, Modbus, CANOpen, CC-Link, etc
- Ethernet TCP, UDP
- Ethernet Automation Protocols
  - Modbus/TCP, Ethernet/IP, Profinet, EtherCAT, etc

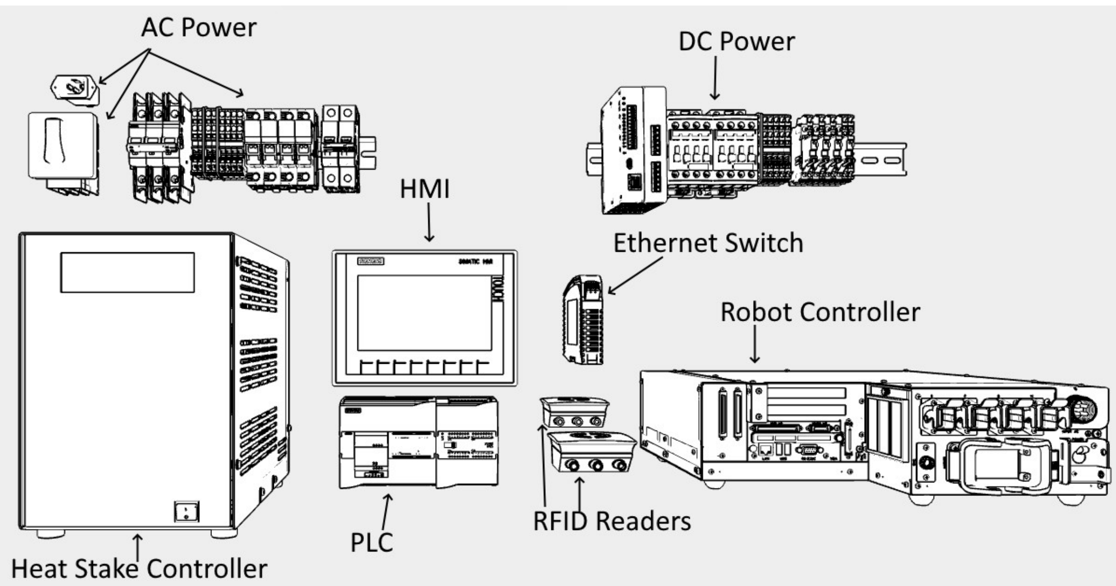
The Details
Component Choices

Major Components – Assembly



- Conveyors
  - Minimize load/unload time
- Robot
  - Quick development
- Nest
  - Precise placement
  - One location simpler
- Heat stake assembly
  - Simplify robot end effector

Major Components - Electrical



- AC Power
- DC Power
- PLC
- HMI
- Robot Controller
- RFID Readers
- Heat Stake Controller
- Ethernet Switch

HSAM Operational Sequence

- Conveyors bring in part trays
  - One conveyor with the bottom, the other with the top
  - The parts can be one of two varieties
- RFID reader reads the trays' tags, sends information to PLC
  - PLC verifies trays match
  - PLC tells robot part type
  - PLC shifts nest to correct location
- Robot removes bottom from tray, places in nest
- Robot removes top from the other tray, places in nest
- Heat stake head moves out and down, performs forming cycle, moves up and back to its rest position.
- Robot places completed part back on the first (bottom) tray

Major Component Choices

Qty	Item	Mfg	P/N	Price	Cost	
1	Power Inlet	Schurter	5120.2007.0	$14.00	$14.00	
1	Disconnect Switch	Lovato	GA016C	$92.00	$92.00	
1	Lockable Operator	Lovato	GAX63	$41.00	$41.00	
1	Circuit Breaker UL489	ABB	SU203M-C10	$150.00	$150.00	
4	Terminal Block	Phoenix	3031393	$2.00	$8.00	ST4 TWIN
4	PE Terminal block	Phoenix	3031416	$7.00	$28.00	ST4 TWIN PE
2	ST4 TWIN Partition	Phoenix	3030789	$1.00	$2.00	ATP-ST-TWIN
1	Circuit Breaker UL1077	ABB	ST202M-C5	$75.00	$75.00	Heat Stake
2	Fuse Holder 2 Pole	TE DL25/18.SFL.2	1SNF100026R0000	$14.00	$28.00	For 10x38 midget
2	Fuse, Midget, 0.8A	LittelFuse	0FLM.800T	$13.00	$26.00	Delta P/S
2	Fuse, Midget, 5.0A	LittelFuse	0FLM005.T	$11.00	$22.00	Robot
				Total	**$486.00**	

Qty	Item	Mfg	P/N	Price	Cost	
1	24V Power Supply	Delta	DRP-24V100W1NN	$82.00	$82.00	3.8A
1	Safety Controller	Banner	SC10-2roe	$600.00	$600.00	Programmable
2	Contactor	Siemens	AF09Z-30-01-21	$75.00	$150.00	DC Use
4	Fuse Block	Phoenix	3036369	$8.00	$32.00	ST4-HESI
1	Fuse 0.63A	Schurter	0000.2502	$2.00	$2.00	Pneumatic
3	Fuse	Schurter	0000.2507	$2.00	$6.00	PLC, Safety
12	Terminal Blocks	Phoenix	3031393	$2.00	$24.00	ST4 TWIN
1	ST4 TWIN Partition	Phoenix	3030789	$1.00	$1.00	ATP-ST-TWIN

Qty	Item	Mfg	P/N	Price	Cost	
1	Robot	Denso Robotics	LPH-040A1	$10,000.00	$10,000.00	
1	Robot Controller	Denso Robotics	RC8A	$5,000.00	$5,000.00	
1	Heat Stake Controller	Amada Weld Tech	UF-4000A	$5,000.00	$5,000.00	
				Total	**$20,000.00**	

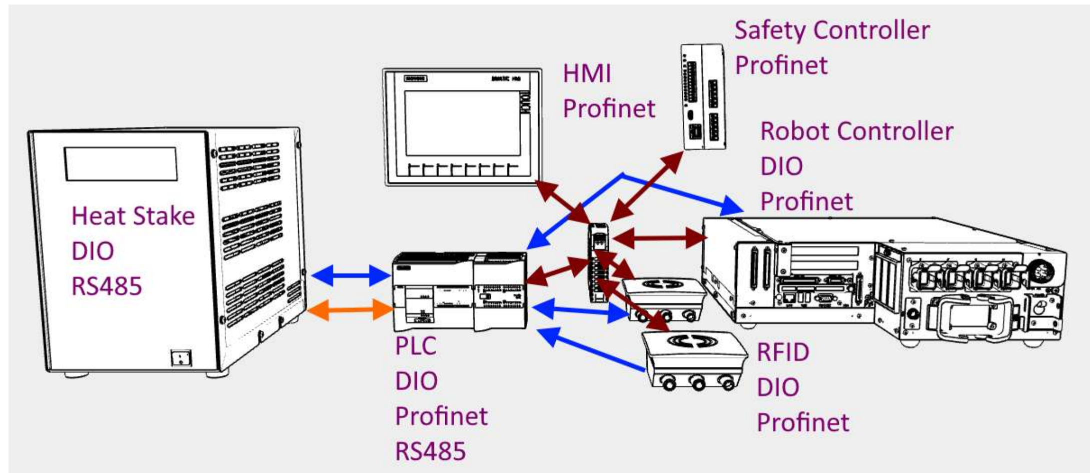
Qty	Item	Mfg	P/N	Price	Cost	
1	PLC 16 DI 10 DO	Siemens	6ES7214-1AG40-0XB0	$450.00	$450.00	S7-1214C
1	RS-485 board	Siemens	6ES7241-1CH30-1XB0	$100.00	$100.00	CB 1241
1	16 DI, 16 DO	Siemens	6ES7223-1BL32-0XB0	$350.00	$350.00	CM 1223
1	HMI	Siemens	6AV21232GB030AX0	$900.00	$900.00	KTP700
				Total	**$1,800.00**	

Qty	Item	Mfg	P/N	Price	Cost	
2	RFID Reader	ifm efector	DTE601	$800.00	$1,600.00	
2	Conveyor			$1,000.00	$2,000.00	
1	Heat Stake Pneumatics			$3,000.00	$3,000.00	
				Total	**$6,600.00**	
				Grand Total	**$29,319.00**	

The Details
Software and Communications

HSAM Communications

- Use Digital I/O for normal operation such as:
  - Triggering RFID read
  - Ask Robot to load next pieces
  - Start Heat Stake cycle
  - Ask Robot to unload part
  - Report error condition
- Use RS485 or Profinet
  - To select setup
  - Report errors
  - Maintenance



Robot Software

- Remove from trays / put back in tray
  - Use palletizing functions
  - Need to teach three points and specify number of rows and columns
- Moving to nest / back from nest
  - Nest position is end point
  - Specify intermediate points for good trajectory
- Tray types
  - Each tray type has its own set of points and variables for palletizing and nest
- Communications with PLC
  - Inputs from PLC: LoadPart, RemovePart, NewTray
  - Outputs to PLC: Ready, TrayDone, Error
  - Profinet from PLC: tray_type integer variable
  - Profinet to PLC: error_code integer variable

Robot Software In Action

- NewTray input ON
  - Robot reads tray_type variable, copies appropriate positions to active point set
  - Robot turns ON Ready output
- LoadNext input ON
  - Runs load routine to place bottom and top on nest
  - Turns ON Ready output
- RemovePart input ON
  - Runs unload routine to place assembled part back onto tray
  - Increments pallet position
  - Turns ON Ready output
- End of tray reached
  - Turns ON TrayDone output
- Error occurs
  - Writes error_code to PLC via Profinet
  - Turns ON Error bit

PLC Software Details 1

- PLC moves everything to Home state
- PLC starts with new trays
  - Turns ON conveyor gate output
  - Turns ON conveyor motors
  - Waits for trays to reach gates (photo electric sensor)
  - Turns OFF conveyor motors
  - Triggers RFID readers
  - Parses RFID code to:
    - Verify bottom and tops match
    - Determinea tray_type
- PLC setup for new tray_type
  - Sends tray_type to robot via Profinet
  - Turns ON NewTray output bit
  - Sends tray_type to Heat Stake Controller using RS485
- PLC assembles next part:
  - PLC turns ON LoadNext for Robot
  - PLC waits for Robot's Ready
  - PLC sets bits to move heat stake mechanism forward and then down
  - PLC triggers heat stake cycle start
  - PLC waits for heat stake Ready signal
  - PLC sets bits to move heat stake mechanism up then backwards
  - PLC turns ON UnloadPart for Robot
  - PLC waits for Ready ON from Robot
  - When Robot turns ON TrayDone
    - PLC turns OFF conveyor gate output
    - PLC starts again with new trays

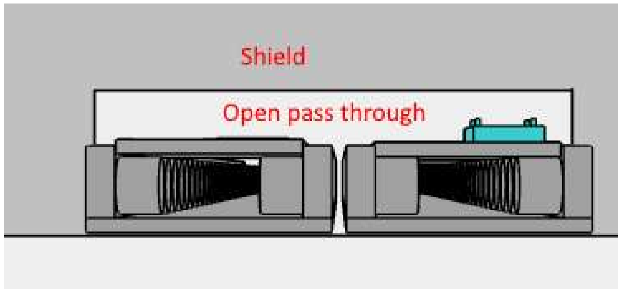
PLC Software Details 2

- Every cycle, PLC checks for:
  - Error signal from Robot
    - If ON, will abort the cycle, read error_code via Profinet and pass to HMI
  - Error signal from Heat Stake
    - If ON, will abort the cycle, read error code from heat stake controller via RS485 and pass to HMI
  - Error signal from Safety Controller
    - If ON, will abort the cycle, read error code via Profinet, and pass to HMI
- PLC code is in a mix of ladder logic and structured text
  - Ladder logic is used for the main sequence, since the code is checking for many conditions such as Cycle Enable, limit sensor status, and time outs
  - Structured text is used for the network communications, since data handling in much easier in structured text
  - Will not use SFC (Sequence Function Charts), because ladder logic and structured text are very standard, but Siemens does not use the standard SFC language.

The Details
Power and Safety

Physical Safety

- Use clear panels to protect the insides of the machine
  - Since there's no manual loading, a light curtain does not make sense
- Access doors at the front and conveyor side
  - Locked during cycle using Solenoid Safety Locks
  - When unlocked, conveyor power is off, robot is in manual (reduced speed) mode. This is handled by safety controller.
  - These locking doors allow quick access for error recovery and maintenance.



- Conveyor safety
  - Conveyor wheels will slip when overloaded
  - Tray pass through opening does not leave space for a hand or arm

AC Power Design 1 - Requirements

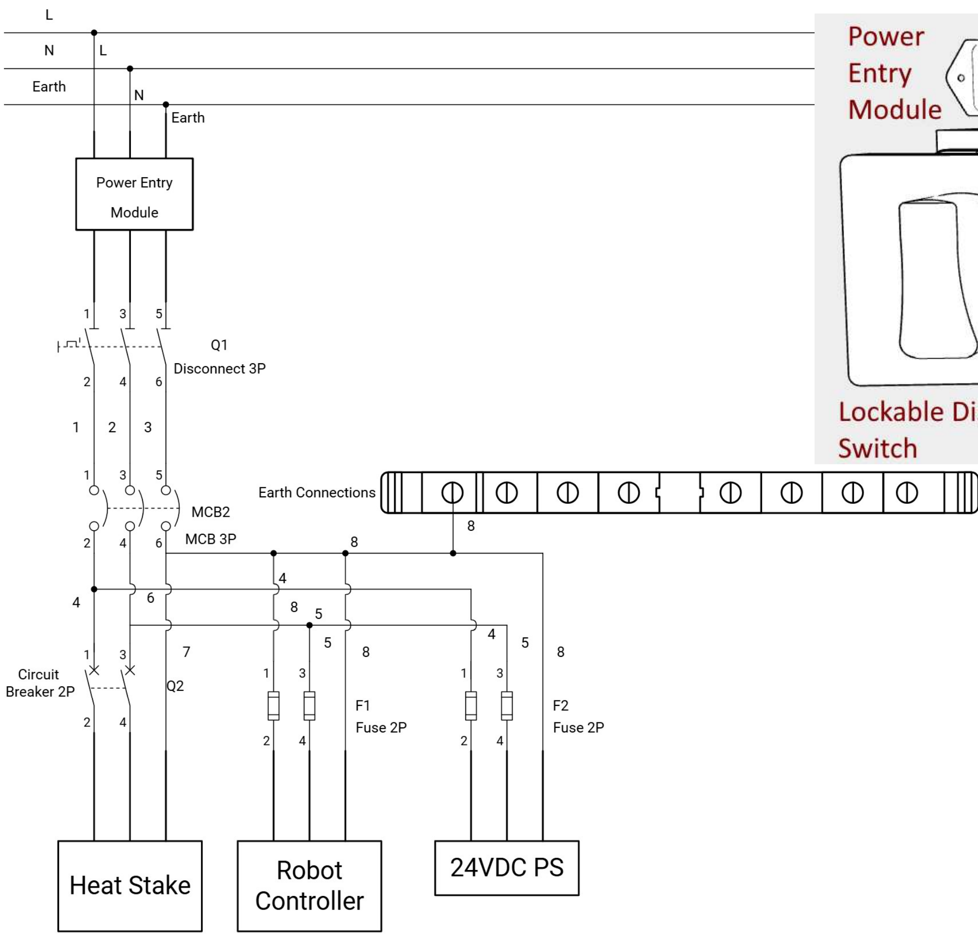
- Three major requirements:

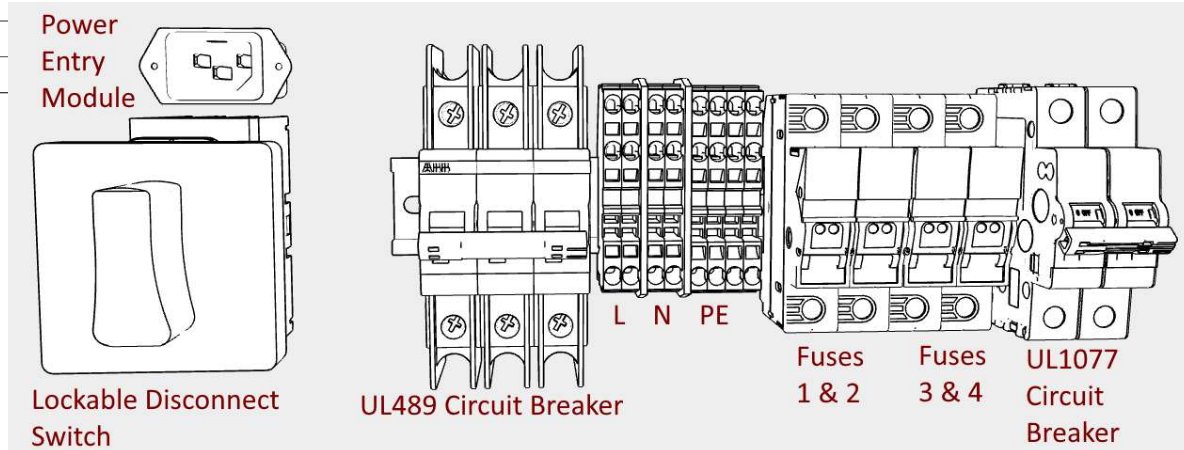
1. SCCR (Short Circuit Current Rating) – machine must be able to handle maximum possible short circuit from the feeder circuit
2. Circuit protection devices to protect against short circuits and over-current
    1. Wires must be sized to handle the maximum current allowed by the protection devices
3. Protect against shock hazards
    1. Electrical components mounted in cabinet rated for the environment.
    2. Inside the cabinet, best practice to use finger-safe components

AC Power Design 2 - Schematic

Schematic and Components

Heat Stake Assembly Machine - AC Power Distribution

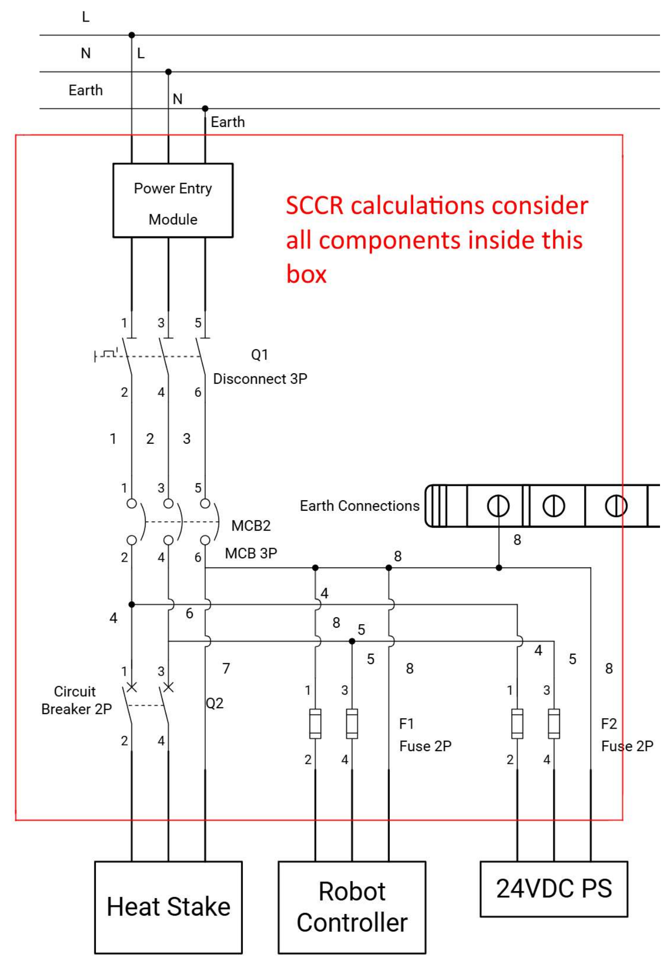




AC Power Design 3 - SCCR

- All the components from the electrical power entry until the last protective component before the load must be considered for SCCR calculations
- The panel's SCCR is the value of the **lowest** rated interrupting capacity
- If an interrupting rating is not specified, then an assumed value from a table is used.
- The HSAM SCCR rating is 10kA:

Component	Interrupting Rating	Notes
Power Entry Module	10kA	Assumed
Disconnect Switch	10kA	
UL489 Circuit Breaker	15kA	
Terminal Blocks	10kA	Assumed
Fuse Holder	100kA	
Fuses	10kA	
UL1077 Circuit Breaker	10kA	

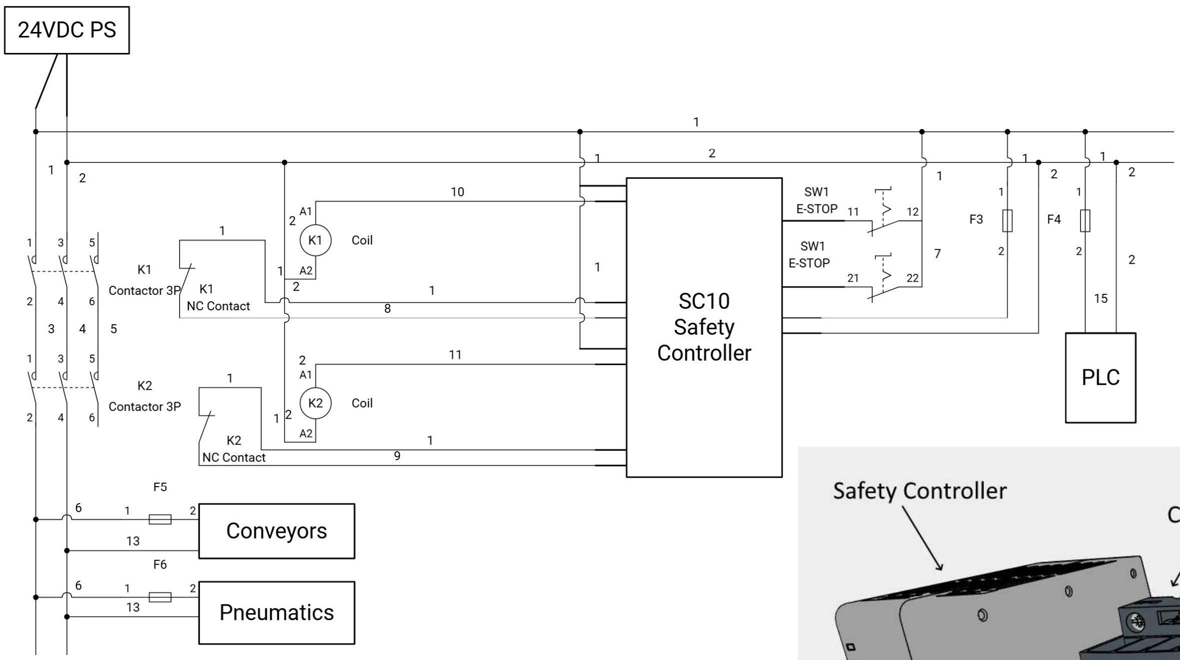


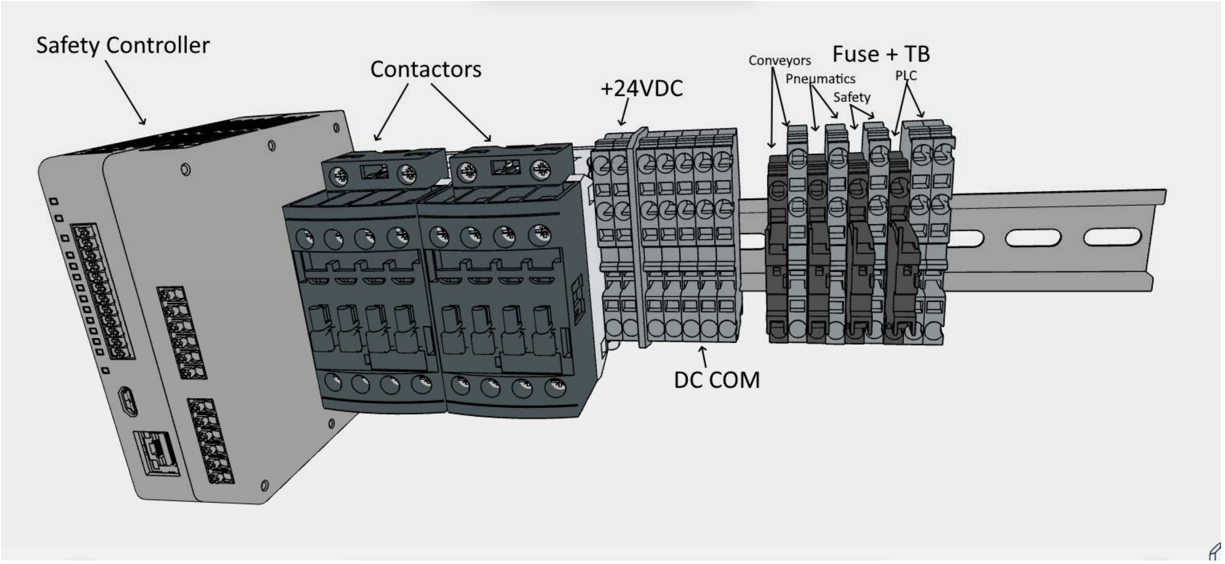
AC Power Design 4 – Protection

- The rated current of the protection device should about 25% more than the maximum continuous current on that branch
- Wiring size is based on maximum continuous current
  - You must follow the guidance from the standard you are using.
  - For example, UL and IEC current ratings are often different.

Branch	Rated Current	125% of Rated	Fuse/CB Size
Heat Stake Controller	4.0 A	5.0 A	5.0 A
Robot Controller	3.5 A	4.4 A	5.0 A
24VDC Power Supply	0.53 A	0.67 A	0.8 A

DC Power Design 1 - Schematic





Summary

Revisiting the Requirements 1

- Safe
  - The machine is designed around typical safety requirements
  - There are no accessible pinch points
- Meet government and customer regulations and certifications
  - The machine is designed to meet NEC and NFPA 79 requirements, could be built to meet UL508A, and could easily be modified to meet SEMI S2
- Be reliable, need minimal operator attention, and be easy to maintain
  - All components are from reputable manufacturers
  - Operator time required is minimal due to conveyor and robot
  - Software is designed with error checking and maintenance in mind
  - Doors are provided with safety locks for rapid but safe access
- Cost effective.
  - Estimated bill of materials is just under $30,000.

Revisiting the Requirements 2

- Needs to be quick to develop – lead customer wants it now!
  - Using robot will speed development time
- Low volume with unpredictable demand
  - Choosing parts that are in stock or have short lead times
  - Using standard terminal blocks
- Customer will be able to maintain machine
  - Using standard PLC languages
  - Siemens and Denso are well known manufacturers
- Accommodate future add-ons such as cameras
  - PLC can add expansion modules
  - Can add more components such as Cognex In-Sight cameras using Profinet